Shrub Mask Generation Pipeline

Technical Review and Statistical Report

Key Messages

- The notebook produced 299 binary shrub masks across 6 sites and saved them as GeoTIFF files aligned to NAIP imagery.
- The mean mapped shrub area per mask is 109.85 m^2, with a range from 14.04 m^2 to 238.32 m^2.
- The reported shrub fraction is exactly 1.0 for every mask. This is a coding and analysis artifact caused by writing background pixels as nodata and then excluding nodata during statistics; it should not be interpreted as 100% shrub cover.
- The relationship between shrub point count and mask area is almost perfectly linear (correlation = 0.999), indicating that output area is driven primarily by the fixed circle-buffer rule used during mask creation.

Executive Summary

This report evaluates the shrub mask workflow implemented in shrub_mask.ipynb and interprets the statistical outputs and figures produced from that workflow. The pipeline is technically coherent and geospatially well structured: it downloads site data, reads shrub coordinate CSV files, applies site-specific transformations, selects ALS tiles, reprojects points into the NAIP reference system, rasterizes buffered shrub locations, and saves the result as compressed GeoTIFF masks.

From a production perspective, the workflow is reproducible and useful for generating point-derived training labels. From an analytical perspective, however, the resulting masks behave more like buffered point labels than true shrub-cover segmentation masks. The evidence is strong: shrub fraction equals 1.0 in every output, background pixel count is always zero in the reported valid area, and shrub area scales almost deterministically with the number of input points.

Therefore, the current workflow is appropriate if the objective is to create standardized binary labels around known shrub centers. It is not yet sufficient if the goal is to quantify real shrub cover, compare occupancy rates between sites, or infer ecological density from the current mask statistics.

Pipeline Review

- The code enumerates six sites, accesses remote directories through WebDAV, and downloads NAIP mosaics, shrub-list CSV files, transformation text files, and ALS metadata.
- For each shrub list, the script detects coordinate columns and an optional radius column. If no valid radius is found, a default shrub radius of 1.0 m is used.
- Shrub coordinates are transformed into ALS space using a homogeneous transformation matrix, then the best ALS tile is selected by counting how many transformed points fall inside each tile's native bounds.
- The points are reprojected into the NAIP raster CRS, a raster window is built around the point cloud with 20 m padding, and a binary mask is initialized.
- Each shrub point is rasterized as a filled circle. The radius in pixels is computed from the metric radius divided by the average NAIP pixel size. The final mask is written as uint8 GeoTIFF with shrub = 1 and nodata = 0.

- A site-level summary table is saved, and a separate deduplication utility removes duplicate masks by site, plot number, and acquisition date.

Dataset Summary

Metric	Value
Number of masks	299
Number of sites	6
Mean width (px)	107.65
Mean height (px)	107.85
Mean shrub pixels	305.14
Mean shrub area (m^2)	109.85
Min shrub area (m^2)	14.04
Median shrub area (m^2)	110.52
Max shrub area (m^2)	238.32
Mean shrub points (n_points)	24.07

Statistical Findings

Across all 299 masks, the average mask width and height are 107.65 px and 107.85 px, indicating that crops are relatively consistent in size.

The mean shrub area is 109.85 m^2, the median is 110.52 m^2, and the maximum reaches 238.32 m^2. The minimum observed shrub area is 14.04 m^2.

The average number of shrub points per mask is 24.07, and the effective mapped area per input point is approximately 4.56 m^2/point.

Among site averages, shaver-lake has the highest mean shrub area (126.61 m^2), followed by sedgwick (122.51 m^2) and calaveras-big-trees (119.06 m^2). independence-lake has the lowest site mean (74.28 m^2), but it also contains some of the largest individual masks, which indicates high within-site variability.

Sedgwick contributes the largest number of masks (83), followed by calaveras-big-trees (73). shaver-lake contributes the fewest (23).

Site Comparison (m^2 unless noted)

Site	Masks	Mean area	Median	Min	Max	Mean n_points	Area/point
calaveras-big-trees	73	119.06	121.32	32.76	237.24	25.99	4.58
dl-bliss	27	98.17	97.56	53.28	167.04	21.81	4.50
independence-lake	56	74.28	59.58	28.08	238.32	16.23	4.58
pacific-union-college	37	115.23	113.40	42.12	207.36	25.24	4.56
sedgwick	83	122.51	126.00	14.04	201.60	26.86	4.56
shaver-lake	23	126.61	124.20	56.16	200.52	27.74	4.56

Interpretation of the Outputs

The near-perfect correlation between n_points and shrub area (0.999) is the most important diagnostic result. It shows that area is being controlled by the rasterization rule rather than by independent variation in shrub geometry. In other words, the label area is almost a deterministic function of point count plus local overlap among circles.

The site-level ratio of mean shrub area to mean n_points is highly stable at about 4.5–4.6 m^2 per point across all sites. This strongly suggests that the default buffering rule is dominating the output. Based on the code and the reported pixel-area statistics, this is consistent with a roughly 2-pixel-radius disk on ~0.6 m NAIP pixels.

The shrub_fraction values of 1.0 for all masks should not be interpreted as complete shrub occupancy. In the current implementation, background pixels are stored with the same value used as nodata (0). When the downstream statistics exclude nodata, only shrub pixels remain in the valid sample, forcing shrub_fraction to 1.0 and bg_pixels to 0.

The gallery plots also support the conclusion that outputs are compact buffered patches centered on shrub coordinates, rather than full scene segmentation products.

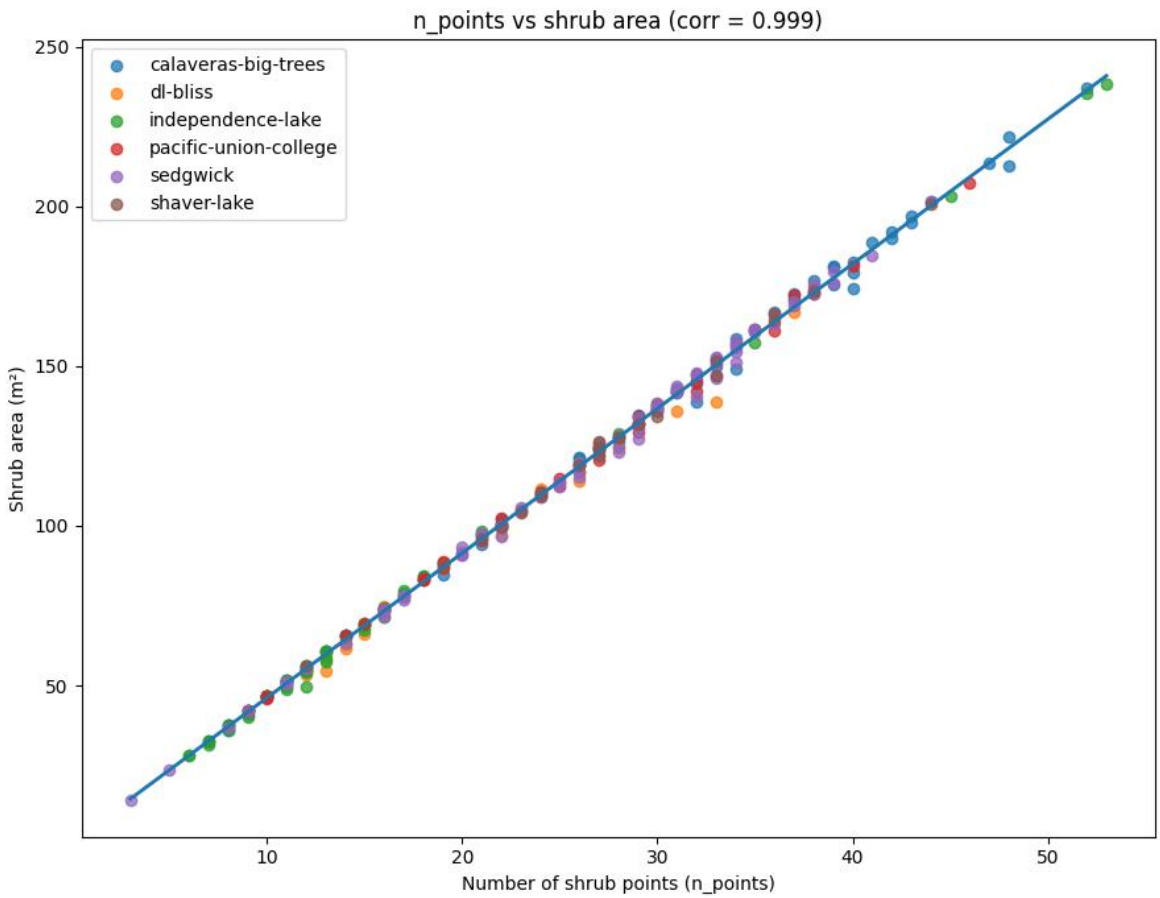

Figure 1. Near-perfect linear relationship between input shrub-point count and mapped shrub area.

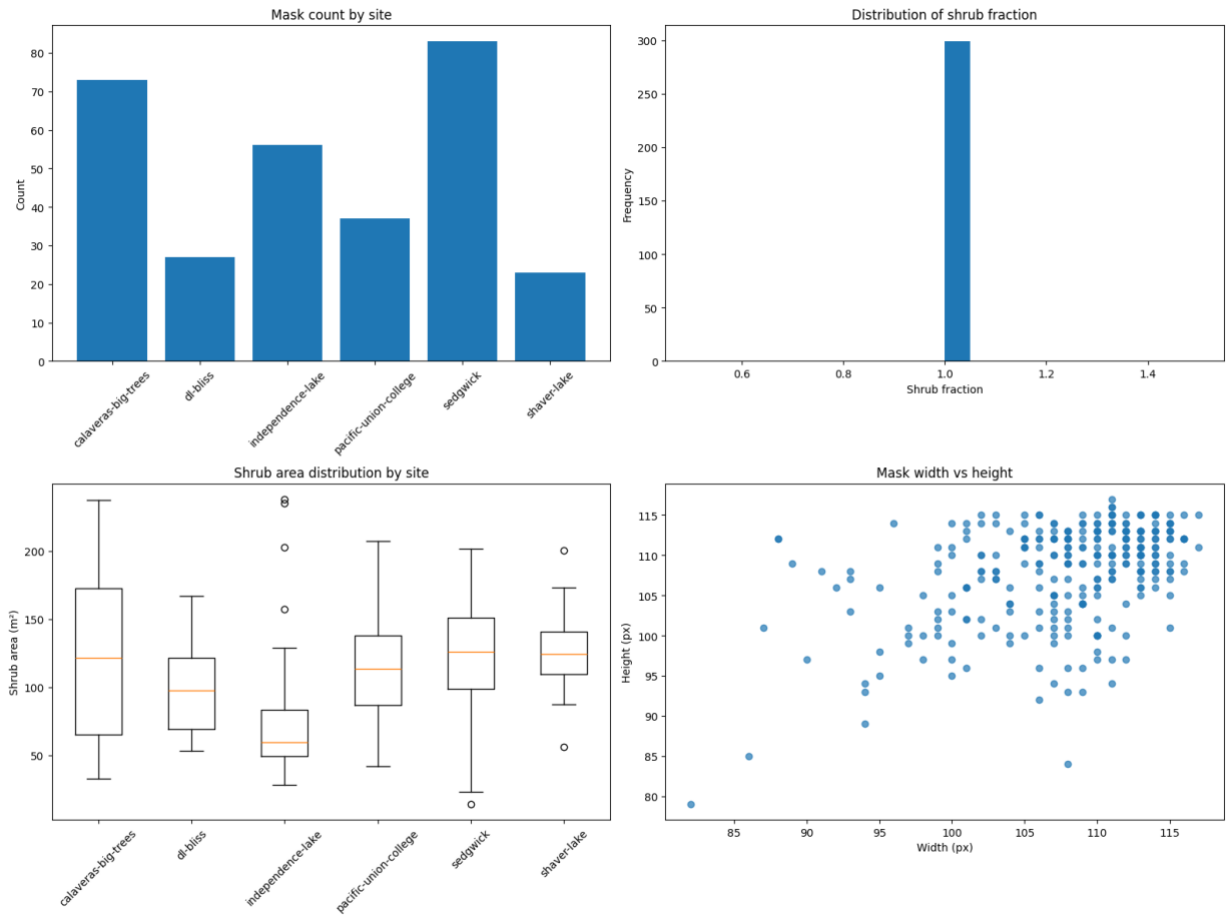

Figure 2. Multi-panel statistical summary: mask counts, shrub-fraction histogram, site boxplots, and width-height scatter.

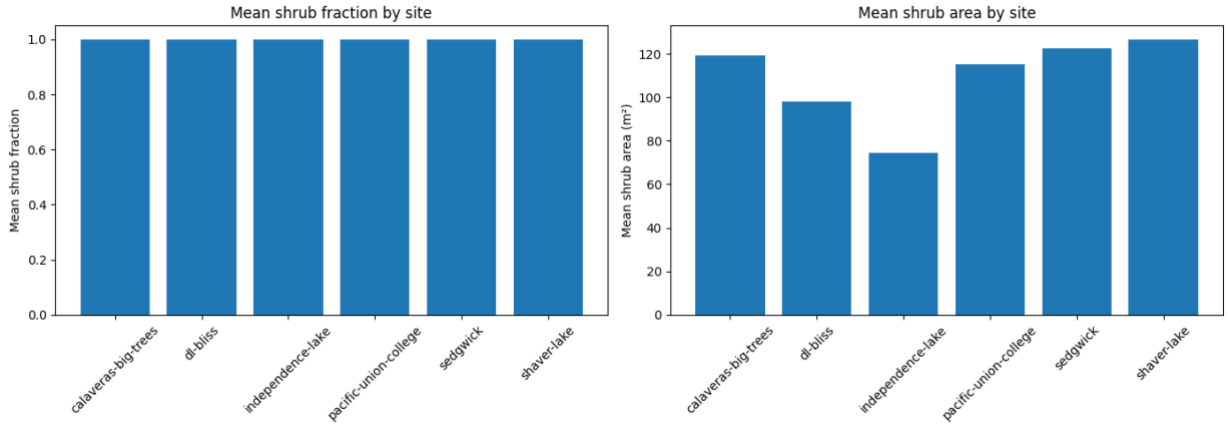

Figure 3. Site-level means. The shrub-fraction bars are all 1.0 because background was encoded as nodata and then excluded from valid-pixel statistics.

Technical Strengths

- Clear end-to-end geospatial workflow with explicit CRS handling and reproducible file outputs.
- Reasonable site abstraction and modular helper functions for downloading, transforming, rasterizing, and saving.
- Use of PDAL metadata for ALS-tile selection is a practical solution for multi-tile environments.
- Compressed GeoTIFF export and summary CSV generation improve reproducibility and downstream usability.
- A visual QA step overlays an example mask on NAIP imagery, which is a good validation practice.

Issues and Risks

- Background and nodata are both coded as 0. This causes downstream fraction and background statistics to become misleading.
- The output behaves as a point-buffer label generator, not as a true segmentation pipeline that estimates shrub footprint from imagery or canopy structure.
- A fixed default radius of 1.0 m may oversimplify shrub size variation across sites and dates.
- Choosing the 'best' ALS tile solely by point count may be brittle near tile edges or in cases where points span multiple tiles.
- Only one example overlay is shown in the notebook. Broader spatial QA would improve confidence.

Largest examples

Site	Mask file	Area (m^2)	n_points
independence-lake	CATNF_6041_20240814_1_mask.tif	238.32	53
calaveras-big-trees	CATCU_0001_20240923_1_mask.tif	237.24	52
independence-lake	CATNF_6050_20240812_1_mask.tif	235.44	52
calaveras-big-trees	CATCU_0043_20241023_1_mask.tif	221.76	48
calaveras-big-trees	CATCU_0081_20250612_1_mask.tif	213.48	47

Smallest examples

Site	Mask file	Area (m^2)	n_points
sedgwick	CASBC_0053_20241117_1_mask.tif	14.04	3
sedgwick	CASBC_0046_20241117_1_mask.tif	23.4	5
independence-lake	CATNF_6049_20240813_1_mask.tif	28.08	6
independence-lake	CATNF_6009_20240731_1_mask.tif	28.08	6
independence-lake	CATNF_6025_20240731_1_mask.tif	31.32	7

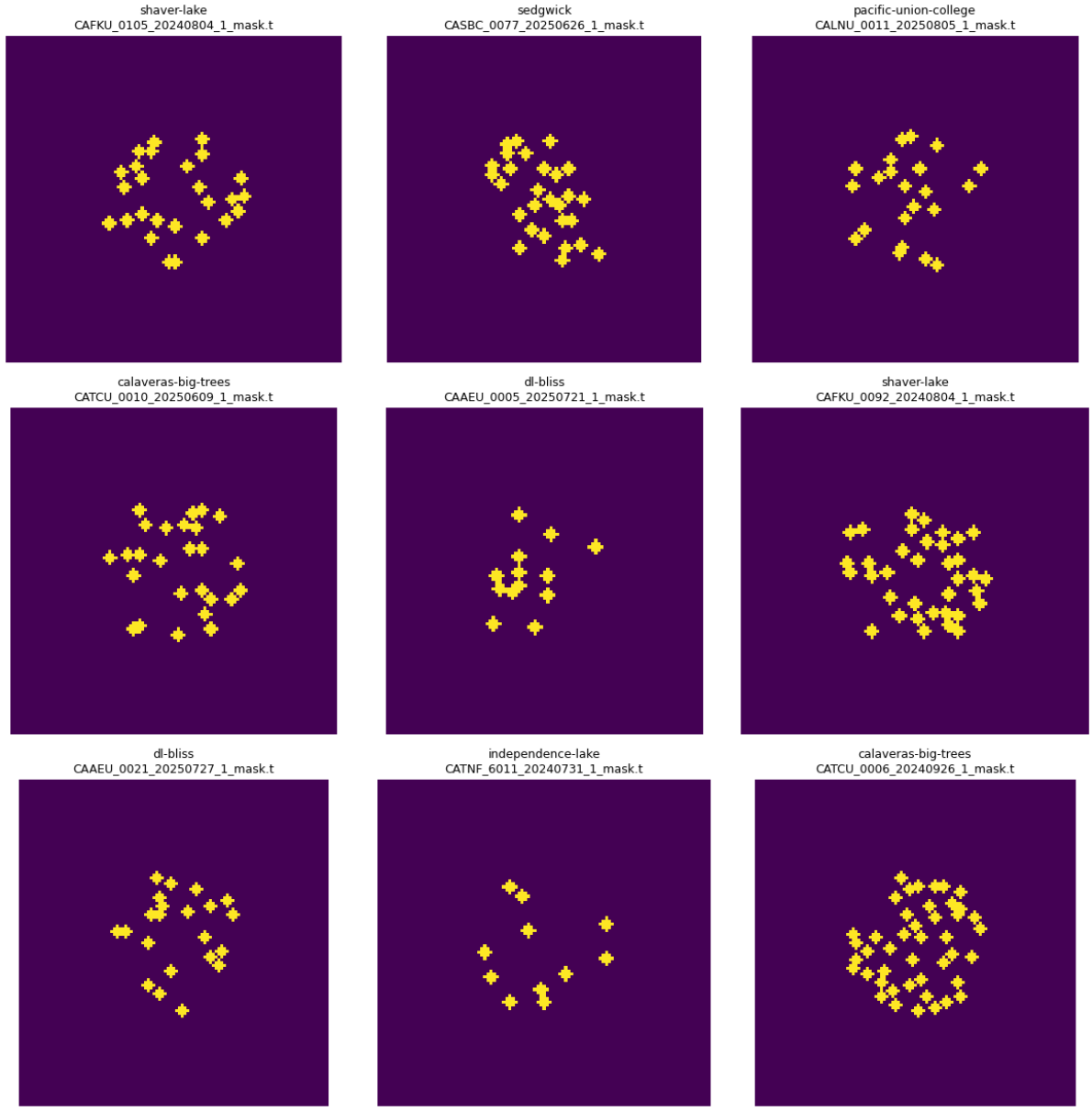

Figure 4. Example mask gallery. The compact buffered shapes around shrub centers are visually consistent with point-buffer labels.

Recommendations

- Separate background from nodata. For example, keep background = 0, shrub = 1, and set nodata = 255. This change will make shrub_fraction and bg_pixels meaningful.
- If the objective is ecological mapping or cover estimation, derive shrub footprint from imagery or canopy products rather than imposing a fixed-radius disk around each point.
- Preserve both point-derived masks and scene windows so that future models can distinguish between label geometry and true image evidence.
- Add formal QA outputs such as per-mask overlays, edge-case logs, and a small random audit sample from every site.
- If variable shrub size is expected, use a validated radius attribute or a site-specific calibration rather than one universal default value.

- Document whether the intended use case is training-label generation, occupancy mapping, or canopy-cover estimation, because the correct mask design differs for each purpose.

Conclusion

The notebook successfully generates georeferenced shrub masks and organizes the results in a reproducible way. Technically, it is suitable for creating standardized binary training labels from shrub-point observations.

However, the current statistics should be interpreted with caution. The apparent 100% shrub fraction is not an ecological result; it is a by-product of the nodata/background encoding. Likewise, the very strong area–point-count correlation shows that the masks mainly reflect the imposed buffering rule.

In summary, the pipeline is strong as a label-generation utility, but it should be revised before being used for quantitative shrub-cover analysis or inter-site ecological comparison.